Science Olympiad — BullSO2026 Div B

Exam Preparation

You will need:

1. Folders for each of the teams to hold the tests
2. Sufficient copies of the test for all teams. They don't need to be stapled.
3. Multiple timers which have a lap function on them - ideally one per volunteer. The timer app on an iPhone or Android Phone that has a stopwatch function with lap function is sufficient.

Before the event begins:

1. Practice starting the timers and using the lap function to record the times. Make sure volunteers understand how to use the lap function and are not accidentally stopping the timer completely.
2. Memorize the answer to the timed question.
3. Check to make sure that this key matches the test you are proctoring.
4. Place one copy of the test for each team in the provided folders with the first page outside the folder.
5. Adjust desks and chairs – teams may have up to 3 students for this event.

Running the Event

1. When the students enter the room, instruct them to sit down, DO NOT OPEN THE FOLDER, and put their names, school name and school number on the first page.
2. Encourage them to write their team number on all the other pages AFTER they begin the test. This way if their papers gets separated from each other we can make sure to give them credit.
3. **CRITICAL:** Check to see that students have ONLY brought
    i. Something to write with (pencils, pens, erasers, highlighters)
    ii. Non-programmable, non-graphing calculators.
    iii. If the student has a smart watch (Apple watch, Samsung Gear, etc.) they will need to put it away.
4. Instruct the students that if they answer the timed question within 10 minutes, they can be awarded a bonus if they solve the timed question with no more than 2 letters incorrect.
    i. When they have a solution for the cryptogram they should raise their hand.
    ii. Let them know that you will announce when the 10-minute time is up. After the first 10 minutes, no additional bonus points will be awarded.
    iii. When you see a team raise their hand, hit the LAP function and head to the team.
    iv. Determine if their answer is correct (see next page for grading), If so, write the time on their score sheet.
    v. If their score is incorrect (more than 2 letters incorrect), tell the team that the answer is wrong, but DO NOT tell them what is wrong. They can continue to work on the question and raise their hand again to be checked. A team has an unlimited number of attempts during the 10-minute bonus.
5. Tell the teams that they do not have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It will not be graded.
6. When the timers hit the 10-minute point, announce that no bonus points will be awarded and put away the timers. The students may continue to work on the question, but they may not receive any extra points.
7. A team is not restricted to only the timed question during the 10 minutes. They can move on or split up the work if they would like, but it is in their best interest to try for the bonus.
8. When time is up, have the students put writing instruments down and put their answer pages back into the folder in the correct order.

How to grade

1. Teams can have up to two incorrect letters total on their cryptogram and still be correct. The frequency of the incorrect letter is irrelevant. See the example below.
   If the cryptogram was as shown:
   KZBAOF KFXMFXYF
   SAMPLE SENTENCE

   and the students answered (underlined letters indicate mistakes)
   SAMPL<u>F</u> S<u>F</u>NT<u>F</u>NC<u>F</u>

   then it counts as four mistakes (even though the mistake was only in the letter E) and the answer DOES NOT count. However, if they put
   SAMP<u>UL</u> SENTENCE

   It is considered correct with two letter mistakes.
2. For Cryptarithms, no "free" mistakes are allowed, and each error will result in a penalty as described in the "As you score each question" section below. These questions are denoted with the symbol ≡ on the scoring sheet as a courtesy reminder for graders.
3. Teams do NOT have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It is NOT to be graded. It is included in the answer key as an aid to the grader.
4. When scoring the Baconian ciphers (with strange text or symbols), they can write the answer under the Baconian symbols or on the line provided. Note that you will see lots of As and Bs, but they are not graded as the answer, only what they put on the answer line.
5. As you score each question, if correct, put the number of incorrect letters (0, 1, or 2) next to the question number on the scoring page. Also, put the value for the question into the score column. If they get more than 2 letters wrong, subtract 100 points from the score until it would be zero. If a question is worth 240 points and they get 4 letters wrong, you would start with 240 points (for up to 2 letters wrong) and then subtract 100 points for each of the next two letters wrong ending up with a final score of 40 points for that question. If they had gotten 5 or more letters wrong on a 240 point question, they would receive 0 points for that question. With a 650 point question, they could get 8 letters wrong and receive 50 points (2 free letters then 6×100=600 points off). Just put the incorrect cost deduction on the score sheet and subtract it from the value for the question. Under no circumstance should the score for any question be less than zero. Note that while the timed question must have 2 or fewer letters incorrect in order to get the timing bonus, a team solving the timed question after the 10 minutes passed would be accepted as correct with 3 incorrect letters receiving 100 points for the timed question.
6. When scoring questions, if they actually did any work on the question and got some letters right, it is useful to write "LOTS" for the question on the score sheet. In this way it is really easy to break ties without having to go look at each question.
7. On a test with Special Bonus Questions (indicated by a ★), if they receive full score for the corresponding question, they will receive an additional bonus depending on the number of special bonus questions they answered. If they answered any one with no deduction points they receive 150 extra points. If they answered any two they receive 400 extra points. If they answered all three then they receive 750 extra points. A legend is at the bottom of the score sheet to make it easy to figure out the extra bonus.
8. If they correctly answered the timed question in 10-minutes or less with 2 or fewer letters incorrect, you need to compute the bonus time. Take the value for the minute from this first table below

0:xx	1,080	1:xx	960	2:xx	840	3:xx	720	4:xx	600
5:xx	480	6:xx	360	7:xx	240	8:xx	120	9:xx	0

   and then add the seconds value from this table:

X:00	120	X:01	118	X:02	116	X:03	114	X:04	112	X:05	110

X:06	108
X:12	96
X:18	84
X:24	72
X:30	60
X:36	48
X:42	36
X:48	24
X:54	12

X:07	106
X:13	94
X:19	82
X:25	70
X:31	58
X:37	46
X:43	34
X:49	22
X:55	10

X:08	104
X:14	92
X:20	80
X:26	68
X:32	56
X:38	44
X:44	32
X:50	20
X:56	8

X:09	102
X:15	90
X:21	78
X:27	66
X:33	54
X:39	42
X:45	30
X:51	18
X:57	6

X:10	100
X:16	88
X:22	76
X:28	64
X:34	52
X:40	40
X:46	28
X:52	16
X:58	4

X:11	98
X:17	86
X:23	74
X:29	62
X:35	50
X:41	38
X:47	26
X:53	14
X:59	2

For example if they solved the time question at the 6:46 mark, you would add 360 (from the 6:xx entry in the first table) to 28 (from the X:46 entry in the second table) to get a bonus of 388. If they had solved it in exactly 4:00 minutes, you would add 600 and 120 to get a bonus of 720.

9. Add up all the scores and put the total on the bottom of score sheet.
10. You must break all ties. Indicate the tie breaker by adding .1 to the score of the team ahead. With multiple teams tied, you will add more. I.e. if five teams all scored 200 points, the final scores that you would enter on the score sheet would be 200.4, 200.3, 200.2, 200.1 and 200.
11. To determine how to break the tie, you need to look at the correctly answered questions in the order from the table below. If both teams answered the same (i.e. they answered the question with zero mistakes) then you go on to the next question. If one team had no mistakes and the other team had one mistake, then the team with no mistakes is ahead. For example, if one team answered question the highest value question and another team didn't, the first team will be ahead. Note that if a test is completely blank, with no evidence of work on any of the ciphers, it should be awarded a score of "P".

Tie Breaker Order	Question #
1	7
2	17
3	3
4	16
5	20
6	Timed
7	19
8	9
9	13
10	14
11	11
12	6
13	4
14	12
15	22
16	15
17	21
18	2
19	5
20	8
21	10
22	1
23	18

12. If there is still a tie (typically when you have teams which answered either zero, one or two questions) then you will need to look at the tie breaker questions again and count the number of correctly answered letters. The team with the most correctly matched letters is to be ahead.

Timed Question **[295 points]** A quote has been encoded using the Aristocrat Cipher for you to decode. When you have solved it, raise your hand so that the time can be recorded and the solution checked. When you have solved it, raise your hand so that the time can be recorded and the solution checked.

PZ NLTHHI YTD'Z GL ZKTZ KTNW ZB CTXL T QBNZI-QBVN
IT REALLY CAN'T BE THAT HARD TO MAKE A FORTY-FOUR

ITNW QPLHW EBTH. IBV WLOLNFL ZB ELZ OHPCLW BVZ TQZLN
YARD FIELD GOAL. YOU DESERVE TO GET SLIMED OUT AFTER

CPOOPDE ZKTZ.
MISSING THAT.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		7	3	2	3	1	1	5	4		3	10		7	4	5	4			10		3	5	1	1	12
Replacement	X	O	M	N	G	V	B	L	Y	W	H	E	P	R	S	I	F	J	Q	A	Z	U	D	K	C	T

★(Special Bonus Question) 1) **[123 points]** Your school's intercom crackled to life with static… then spelled out this Affine message by Benjamin Franklin with ***a*=21** and ***b*=10**. What does it say?

E	Q	H	H	V	S	X	Q	W	Y	F	Q	T	T	Q	D	T	B	K	X	E	Q	H	H	Y	K	W	V	.
W	**E**	**L**	**L**	**D**	**O**	**N**	**E**	**I**	**S**	**B**	**E**	**T**	**T**	**E**	**R**	**T**	**H**	**A**	**N**	**W**	**E**	**L**	**L**	**S**	**A**	**I**	**D**	.

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2) **[209 points]** You open a forgotten email draft saved in the school's computer system. It holds a quote encoded in a Complete Columnar. The quote has **BUBBLES** somewhere in it. What does it say?

ORSLO HOAIR XFSBW MIAIO TXHEE UESDN WAXHU NOTEA

YONXI TBERR SSPUG TWLOG CNOPW XGBIF EPCLH IXUEM

DMACL TNX

Answer:
IF THOUGHTS WERE BUBBLES, MINE WOULD FORM GEOMETRIC SHAPES AND OCCASIONALLY POP WITHOUT WARNING.

3) **[447 points]** A vending machine just dispensed a candy wrapper with this Fractionated Morse code by Goethe. You are told that the quote has **WHAT** in it corresponding to the encoded text **A Q I Y P**, meaning **A** = ●—; **I** = ●●×; **P** = –××; **Q** = ×●●; **Y** = ●–×. What does it say?

```
 Y  V  S  U  X  I  B  Q  E  I  J  W  F  B
●–××––×●–×–●××●●●×●×●×●●●××●●×–●××–×●●●●×●
A / M  A  N / S  E E S  / I  N / T H  E

 X  M  L  S  B  G  U  E  A  Q  I  Y  P  F  B
××●––×–––×●–●×●–●●×–●●×××●––×●●●●×●–×–××●●●●×●
/ W  O  R  L  D  / W  H  A  T/ H  E

 Z  D  S  S  B  J  I  B  I  Q  U  X  F  Q  Q  E
××–●–●×●–×●–●×●–●×●●×●×●●●××●●×–●××●●●●×●●×●●●××
/ C  A  R  R  I  E S  / I  N / H  I  S  /

 F  B  S  S  C
●●●●×●×●–×●–●×–
H  E A  R  T
```

Replacement	F	R	I	D	A	Y	B	C	E	G	H	J	K	L	M	N	O	P	Q	S	T	U	V	W	X	Z
Morse fraction	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

4) **[243 points]** A fragment of text is discovered in a time capsule from 1965. It's a quote encrypted as a Checkerboard Cipher with a polybius key of **CUMBERLAND**. What does it say?

HR	EN	EP	CT	CP	HI	HN	AP	HN	HR
L	I	F	E	C	A	N	O	N	L

PN	CN	CT	CR	HN	HT	CT	HP	AN	AT
Y	B	E	U	N	D	E	R	S	T

AP	AP	HT	CN	HI	CP	ET	PR	HI	HP	HT	AN	CN	CR	AT	EN	AT	CI	CR	AN
O	O	D	B	A	C	K	W	A	R	D	S	B	U	T	I	T	M	U	S

AT	CN	CT	HR	EN	PP	CT	HT	EP	AP	HP	PR	HI	HP	HT	AN
T	B	E	L	I	V	E	D	F	O	R	W	A	R	D	S

5) **[206 points]** A riddle is discovered taped under a chemistry lab bench. It's a quote by Francois de La Rochefoucauld, encrypted as an Aristocrat. What does it say?

U STJY CTAYFE AK SOY NPKS BTYXAPJK PC UDD
A TRUE FRIEND IS THE MOST PRECIOUS OF ALL

BPKKYKKAPFK UFE SOY PFY QY SUMY SOY DYUKS SOPJZOS
POSSESSIONS AND THE ONE WE TAKE THE LEAST THOUGHT

ULPJS UXRJATAFZ.
ABOUT ACQUIRING.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	6	2	2	3	2	5				5	9	1	1	1	5	8	1	1	10	4	7			2	11	2
Replacement	I	P	F	L	D	N	V	X	Z	U	S	B	K	M	H	O	W	Q	T	R	A	J	Y	C	E	G

6) **[245 points]** While debugging a program, you find this hidden in the comments. It's a quote by John Marshall, encrypted with a Complete Columnar. The quote has **LISTEN** somewhere in it. What does it say?

LWARE CINFE ALTEI WAOUI INOEI ESFAO CALAL XSLPU

NMANU SKXON SEMFN ONCTL TLOLS MTDET WX

Answer:
TO LISTEN WELL IS AS POWERFUL A MEANS OF COMMUNICATION AND INFLUENCE AS TO TALK WELL.

—

7) **[532 points]** Decrypt the following cipher text that has been encoded using the Patristocrat Cipher using a K1 alphabet. What does it say?

QKIFA DNPBQ KYHNC JBWIC QAUPI HNYYI CWIWZ PNWXH
THEAL GORIT HMCON FIDEN TLYRE COMME NDEDP RODUC

QEGFE IWNCY UGPNS EBCDK BEQNP USKBH KYNEQ AUPIJ
TSBAS EDONM YBROW SINGH ISTOR YWHIC HMOST LYREF

AIHQE GNPIW NYFCW YBAWH NCJXE BNC
LECTS BORED OMAND MILDC ONFUS ION

The algorithm confidently recommended products based on my browsing history, which mostly reflects boredom and mild confusion.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	7	8	2	7	3	3	6	9	3	5			12		7	7		2		4		8	2	7	1
Replacement	L	I	N	G	S	A	B	C	E	F	H	J	K	O	Q	R	T	V	W	X	Y	Z	D	U	M	P

★(Special Bonus Question) 8) **[205 points]** Someone left this coded message in the school lab. You are told it's a quote by Marcus Aurelius encoded with a Nihilist Substitution with a keyword of **JULIEN** and polybius key of **ABOLISHMENT**. What does it say?

29	58	35	36	39
L	**O**	**S**	**S**	**I**

46	40	58	45	37
S	**N**	**O**	**T**	**H**

39	50	50	69	28
I	**N**	**G**	**E**	**L**

36	48	37	60	76
S	**E**	**B**	**U**	**T**

46	37	35	50	50
C	**H**	**A**	**N**	**G**

69	25	40	57	57
E	**A**	**N**	**D**	**C**

37	56	39	50	48
H	**A**	**N**	**G**	**E**

40	36	70	25	46
I	**S**	**N**	**A**	**T**

69	69	39	66	47
U	**R**	**E**	**S**	**D**

39	38	40	50	67
E	**L**	**I**	**G**	**H**

45
T

9) **[259 points]** Found inside a dissected frog's stomach (plastic, don't worry): this Aristocrat note. You are told that the cipher **starts with the word "The"**. What does it say?

ZSY MPKJIDZUFUZA EIPI QRPTYJ CY ROKIZ JUVZPRDZUKTV.
THE PRODUCTIVITY GURU WARNED ME ABOUT DISTRACTIONS.

U UCCYJURZYGA VIOVDPUOYJ ZK ZSPYY CKPY TYQVGYZZYPV.
I IMMEDIATELY SUBSCRIBED TO THREE MORE NEWSLETTERS.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2		4	3	1	1	2		5	5	5		1		3	8	2	4	2	3	8	6			12	11
Replacement	Y	F	M	C	G	V	L	X	U	D	O	Z	P	Q	B	R	W	A	H	N	I	S	J	K	E	T

10) **[145 points]** You uncover this encoded note tucked behind a microscope slide. It's a quote by Lily Tomlin, written in a Porta using a keyword of **ADVISOR**. What does it say?

A	D	V	I	S		O	R	A	D	V		I	S	O	R	A		D	V	I	S	O		R	A	D	V	I		S	O	R	A	D
V	O	V	F	W		E	K	J	O	D		C	N	X	L	B		P	O	B	F	T		Z	O	A	N	H		X	A	L	V	E
I	**A**	**L**	**W**	**A**		**Y**	**S**	**W**	**A**	**N**		**T**	**E**	**D**	**T**	**O**		**B**	**E**	**S**	**O**	**M**		**E**	**B**	**O**	**D**	**Y**		**B**	**U**	**T**	**I**	**S**

V	I	S	O	R		A	D	V	I	S		O	R	A	D	V		I	S	O	R	A		D	V	I	S	O
R	K	L	S	Y		U	O	L	V	X		Y	Z	A	N	E		A	N	L	H	R		Q	S	W	R	W
H	**O**	**U**	**L**	**D**		**H**	**A**	**V**	**E**	**B**		**E**	**E**	**N**	**M**	**O**		**R**	**E**	**S**	**P**	**E**		**C**	**I**	**F**	**I**	**C**

11) **[245 points]** Someone left this encrypted quote scribbled on the chalkboard before class began. It's a quote, written as an Aristocrat. What does it say?

DPRZ DEAX VR DE DWFOD DPR CWEBROO. DPR CWEBROO IEK
THEY TOLD ME TO TRUST THE PROCESS. THE PROCESS NOW

PMO VR HMAMIBUIL OCEEIO QEW IE BARMW WRMOEI.
HAS ME BALANCING SPOONS FOR NO CLEAR REASON.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	4	3	7	10	1		1	6		1	1	5		9	4	1	9			1	2	6	1		1
Replacement	L	C	P	T	O	U	Z	B	N	K	W	G	A	Q	S	H	F	E	X	V	I	M	R	D	J	Y

12) **[237 points]** Solve this Baconian by Byron Pulsifer. You are told that the deciphered text starts with **Fate**.

```
BIRDS ABIDE PHASE CARLA BRIDE NICKY A WALK LOYAL
AABAB AAAAA BAABA AABAA ABAAA BAAAB ABAAA  ABBAA

  F     A     T     E     I     S     I     N


OF ONE JONES OCCUR OBAMA ABOUT AGILE ARNIE ADIOS
BABBA  ABBAB BAABB BAAAA AABBB AAAAA ABBAA AAABB

  Y     O     U     R     H     A     N     D


REIGN ADELE JORGE ABBOT A ROAD ATOMS BONES ANNIE
BAAAB AAAAA ABBAA AAABB ABBAA  ABBAB ABBAB ABBAA

  S     A     N     D     N     O     O     N


AHOLD CARLA ANDRE NICKY ALPHA SHEEP
AABAA AABAA ABABA BAAAB AABAA BAAAB

  E     E     L     S     E     S
```

Fate is in your hands and no one elses

13) **[256 points]** A laser-etched message on the inside of a geodes crystal reads: solve this Aristocrat version of Ralph Emerson's wisdom. What does it say?

IYAMH MYA HJAR PJD VAA HJMH VSWYWHQMT WV VHYDXIAY
GREAT ARE THEY WHO SEE THAT SPIRITUAL IS STRONGER

HJMX MXR UMHAYWMT GDYLA, HJMH HJDQIJHV YQTA HJA
THAN ANY MATERIAL FORCE, THAT THOUGHTS RULE THE

PDYTN.
WORLD.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	10			5			1	13	3	8		1	9	1		2	3	2	1	4	1	5	4	3	9	
Replacement	E	K	J	O	Z	B	F	T	G	H	V	C	A	D	Q	W	U	Y	P	L	M	S	I	N	R	X

14) **[255 points]** Archaeologists unearth a stone tablet inscribed with a message. It's a quote encoded as a Cryptarithm. What do the values **123423506 7586** decode to?

Values to decode for solution

1	2	3	4	2	3	5	0	6		7	5	8	6
C	O	R	P	O	R	A	T	E		F	A	M	E

Cryptarithm formula

```
  E F F O R T
  6 7 7 2 3 0
+ P R O J E C T
  4 3 2 9 6 1 0
-------------
A T T E M P T
5 0 0 6 8 4 0
```

EFFORT+PROJECT=ATTEMPT

	0	1	2	3	4	5	6	7	8	9
A										
C										
E										
F										
J										
M										
O										
P										
R										
T										

15) **[218 points]** Scrawled in dry-erase marker on the whiteboard after hours: this Aristocrat encoded quote by Helen Keller. What does it say?

QPL GJYQ OLEZQFMZV QPFBIY FB QPL DJKVC HEBBJQ OL
THE MOST BEAUTIFUL THINGS IN THE WORLD CANNOT BE

YLLB JK LXLB QJZHPLC. QPLT GZYQ OL MLVQ DFQP QPL
SEEN OR EVEN TOUCHED. THEY MUST BE FELT WITH THE

PLEKQ.
HEART.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		6	2	2	3	4	2	2	1	5	3	14	2		3	8	13			1		3		1	4	4
Replacement	P	N	D	W	A	I	M	C	G	O	R	E	F	X	B	H	T	J	Q	Y	K	L	Z	V	S	U

★(Special Bonus Question) 16) **[332 points]** The message below was encoded into the DNA sequence of a synthetic organism. It translates to a quote by Buddha, encrypted using a Checkerboard Cipher with a polybius key of **CONSTABLE**. What does it say?

AR AS EH TR SS SH GE TO TE SR

I	**F**	**W**	**E**	**C**	**O**	**U**	**L**	**D**	**S**

TR TR SE AO TR GS AR GR TS SS

E	**E**	**T**	**H**	**E**	**M**	**I**	**R**	**A**	**C**

TO TR SH AS TS SR AR SO AH TO TR AS TO SH EH TR GR SS TO TR

L	**E**	**O**	**F**	**A**	**S**	**I**	**N**	**G**	**L**	**E**	**F**	**L**	**O**	**W**	**E**	**R**	**C**	**L**	**E**

TS GR TO ER SH GE GR EH AO SH TO TR TO AR AS TR EH SH GE TO

A	**R**	**L**	**Y**	**O**	**U**	**R**	**W**	**H**	**O**	**L**	**E**	**L**	**I**	**F**	**E**	**W**	**O**	**U**	**L**

TE SS AO TS SO AH TR

D	**C**	**H**	**A**	**N**	**G**	**E**

17) **[523 points]** Hidden in the source code of a retro video game, this Patristocrat puzzle reveals a quote by Nelson Mandela using a K1 alphabet. You are told that the cipher contains **`the word "altered"`**. What does it say?

```
RYKQK CXDZR YCDEF CBKQK RSQDC DERZH OFHAK RYHRQ
THERE ISNOT HINGL IKERE TURNI NGTOA PLACE THATR

KNHCD XSDAY HDEKJ RZLCD JRYKU HGXCD UYCAY GZSGZ
EMAIN SUNCH ANGED TOFIN DTHEW AYSIN WHICH YOUYO

SQXKF LYHTK HFRKQ KJ
URSEL FHAVE ALTER ED
```

There is nothing like returning to a place that remains unchanged to find the ways in which you yourself have altered.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	1	8	9	3	4	3	8		3	12	2		1	1		6	9	4	1	2			4	8	5
Replacement	C	K	I	N	G	L	Y	A	B	D	E	F	J	M	P	Q	R	T	U	V	W	X	Z	S	H	O

18) **[105 points]** Your solar-powered calculator started flashing this Caesar sequence—it's Brian Tracy's words. What does it say?

G	W	C		P	I	D	M		E	Q	B	P	Q	V		G	W	C	,		Z	Q	O	P	B		V	W	E	,
Y	**O**	**U**		**H**	**A**	**V**	**E**		**W**	**I**	**T**	**H**	**I**	**N**		**Y**	**O**	**U**	,		**R**	**I**	**G**	**H**	**T**		**N**	**O**	**W**	,

M	D	M	Z	G	B	P	Q	V	O		G	W	C		V	M	M	L		B	W		L	M	I	T		E	Q	B	P
E	**V**	**E**	**R**	**Y**	**T**	**H**	**I**	**N**	**G**		**Y**	**O**	**U**		**N**	**E**	**E**	**D**		**T**	**O**		**D**	**E**	**A**	**L**		**W**	**I**	**T**	**H**

E	P	I	B	M	D	M	Z		B	P	M		E	W	Z	T	L		K	I	V		B	P	Z	W	E		I	B		G	W	C	.
W	**H**	**A**	**T**	**E**	**V**	**E**	**R**		**T**	**H**	**E**		**W**	**O**	**R**	**L**	**D**		**C**	**A**	**N**		**T**	**H**	**R**	**O**	**W**		**A**	**T**		**Y**	**O**	**U**	.

19) **[273 points]** You intercepted this transmission from a rogue weather balloon. Decode the Aristocrat version of Byron Pulsifer's famous words. What does it say?

CA UR CBAXMBCNXJ DQZ FGQZ AQJS CDFRV D NRO VRYAQZV
TO BE THOUGHTFUL AND KIND ONLY TAKES A FEW SECONDS

YAKHDERZ CA CBR CGKRJRVV BXEC YDXVRZ US AQR EXZR
COMPARED TO THE TIMELESS HURT CAUSED BY ONE RUDE

MRVCXER.
GESTURE.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	7	4	9	5	4	2	2	1		3	2		2	2	1		5	13	2		2	7		6	3	6
Replacement	O	H	T	A	R	K	I	P	V	L	M	J	G	F	W	X	N	E	Y	Z	B	S	Q	U	C	D

20) **[321 points]** A message flashes briefly on a malfunctioning computer screen. It turns out to be a quote written in an Aristocrat using a K1 alphabet. What does it say?

IS AICMNQCNEB XFNFENFX ANMFAA. NBF ANMFAA QCA IVANWS
MY SMARTWATCH DETECTED STRESS. THE STRESS WAS MOSTLY

YMVI MFEFZPZJU EVJANCJN CWFMNA CDVON IS ANMFAA.
FROM RECEIVING CONSTANT ALERTS ABOUT MY STRESS.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	14	2	6	1	4	10			5	3			7	13	1	1	2		3		1	4	2	2	1	2
Replacement	S	H	A	B	C	E	J	K	M	N	P	Q	R	T	U	V	W	X	Y	Z	G	O	L	D	F	I

21) **[216 points]** Your graphing calculator won't stop displaying this Aristocrat puzzle by Thomas Carlyle using a K1 alphabet. What does it say?

PE MOAKGH WOM SOAHI JSOMNH; IHVQXW CLVV SH ZKLA SLI
BY NATURE MAN HATES CHANGE; SELDOM WILL HE QUIT HIS

XVQ SXWH ALVV LA SOI OJAKOVVE ROVVHM OGXKMQ SLI HOGI.
OLD HOME TILL IT HAS ACTUALLY FALLEN AROUND HIS EARS.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	6		1		2		3	8	6	2	4	6	5	1	10	1	3	1	7			10	3	4		1
Replacement	T	V	W	X	Y	Z	R	E	S	C	U	I	N	G	A	B	D	F	H	J	K	L	M	O	P	Q

22) **[227 points]** A note falls out of a borrowed library book. The message is a quote by Anne Frank, written as an Aristocrat Cipher. What does it say?

YJS SJOWQIUKF NB NA BYVB OJZJWR OQQW SVNB V ANOMFQ
HOW WONDERFUL IT IS THAT NOBODY NEED WAIT A SINGLE

DJDQOB ZQUJIQ ABVIBNOM BJ NDXIJGQ BYQ SJIFW.
MOMENT BEFORE STARTING TO IMPROVE THE WORLD.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	9		3		3	1		5	9	1		2	6	6		9	1	4		2	4	4	1	3	2
Replacement	S	T	J	M	Z	L	V	X	R	O	U	C	G	I	N	K	E	Y	W	Q	F	A	D	P	H	B

BullSO, University of South Florida - Repaired practice edition - Answer key

Division B | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

reasoning: A gives 100 x 0.9 / 3 = 30 expected points per minute.
B gives 200 x 0.3 / 6 = 10 expected points per minute.
Estimates are uncertain; revise them if new constraints make B easier or actual scoring differs. (3 points).

conclusion: A is a reasonable initial priority under these assumptions; the ratio is a planning heuristic, not a score guarantee. (3 points).